

Deploy and onboarding Infrastructure into PCD SaaS control plane



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Prepared By	Arvind Sridharan (Solution Architect)
Customer	NTT
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Objective

- Implement Platform9 PCD in a production-like datacenter environment as a Canary deployment.
- Design and build the OpenStack platform to host VMs in one physical datacenter with a SaaS-based management plane.
- Utilize this environment as the testing, validation, and systems integration test bed for all subsequent production deployments.
- Limit the Canary environment to no more than four (4) hypervisors.

Solution Architecture Overview

PCD Deployment

The target architecture is a PCD SaaS control plane hosted in the AWS cloud in the nearest location to the customer's data center. The SaaS Plane will host the critical control components of openstack such as keystone, nova, neutron, cinder, glance and various other control components that will enable to configure and onboard hosts residing in the datacenter which will be used to create PCD cluster that will be hosting the VM/application workloads. The control plane provides the abilities to :

- Integrate AD/entraID SSO login for the Users to connect.
- Onboard and Add/remove Baremetal hosts with specific roles (hypervisor/storage/image) with the PCD clusters
- Integrate and configure the AFF/solidfire/Pure storage drivers for the cinder block store services
- Setup the host network specific profile to suit the requirements.
- Setup Guest networking topology to fulfil the connection requirements
- Configure and manage VM images via glance service.
- Deploy VM workloads using various methods and connect to the existing datacenter fabric.

VM workload migration

Post the PCD cluster is ready and fully functional as per the specification, The existing VMware workloads can be further migrated from the Esxi hosts to PCD hosts using the Vjailbreak tool. This tool provides the ability to:

- Lift and shift the VMs from VMware to PCD clusters.
- Ensure the connectivity/communication of the VM is intact before and after the migration.
- Post a successful migration power provides the ability to take action like retaining the existing VM in power off state.
- Setup schedule to perform the migration on a specific time frame.

Business case

The primary business drivers are to remove VMware licensing risk and lock-in, reduce TCO, and gain strategic control over the virtualization stack with PCD which is built on top of widely accepted open source technology stack like KVM, openvswitch and openstack which also brings in the ability to automate the operations through the openstack APIs which PCD architecture has 100% compatibility with.

Requirement Summary

Business requirements

- Provide functionally equivalent or better VM/baremetal host lifecycle and availability capabilities.
- Allow phased migration of existing workload of VMs from VMware with controlled downtime and clear rollback option using the Platform Vjailbreak tool.

Technical requirements

- Setup MaaS service in the canary environment for imaging/re-imaging baremetal host with ubuntu 24.04 LTS release.
- Onboard and Add/remove Baremetal hosts with specific roles (hypervisor/storage/image) with the PCD clusters
- Integrate AD/entraID SSO login for the Users to connect.
- Integrate and configure the FAS/solidfire/Pure storage driver for the cinder block store services
- Setup the host network profiles to match the requirements for routing VM/app traffic between the host and the data center fabric
- Setup Guest networking topology to match the current architecture implemented for VMware guests.
- Configure and manage VM images via glance service.
- Verify the prometheus metrics exposed from each host which can be used further for observability.
- Setup and configure Vjailbreak tool to perform VM migrations between VMware to PCD clusters.

Infrastructure requirements

Hardware

Baremetal host	Configuration	Network/FC interfaces	Console Access type
Dell M640	Minimum: 8CPU 16GB RAM 250GB disk	5x NICs	IDRAC 9
HP proliant DL360 gen10 plus	Minimum: 8CPU 16GB RAM 250GB disk	5x NICs	iLO5

At this point only x86_64 based processor hardware is supported with PCD releases.

Configurations

- IPMI/redfish service to be enabled on the BMC level
- It should have one **suitable interface to be used for PXE booting** it via MaaS.
- Ensure the hardware CMDB (configuration Management Databases) for the infrastructure are updated with respective configuration/settings as needed for the hosts to be added in PCD

NOTE: For Using Redfish service on Dell server ensure a user name "**maas**" is created with operator privileges in the iDRAC which will be used by MaaS to power cycle the host during installation.

Network:

The required interfaces on the hosts to be wired to the datacenter switch fabric as per the requirements.

Configurations

- A set of free IP addresses which will be used on the hosts for
 - External communication.
 - Storage VLANs
 - VM provider overlay networkto be carved out from the existing network space OR a new segment which is reachable to the existing VMware cluster network should be created.
- Required VLANs to be used has to be defined before onboarding of baremetal
- **For Onboarding baremetal at scale using MaaS the hosts should have a dedicated vlan or native VLAN to pxe boot via the directions of MaaS server**
- **Ensure port 443 is allowed in the environment firewall and proxy for onboarding and connections between Dataplane and the AWS control plane**

Software and platform

- PCD control plane to be deployed with **October Patch-V1** release.
- Baremetal hosts will be installed and configured with **Ubuntu24.04 LTS** release.
- NTP or timesyncd service to be running on the hypervisor nodes.
- Other openstack specific packages to be installed depending on the roles applied to specific baremetal host at the time of creating PCD cluster

Maas Setup and configuration

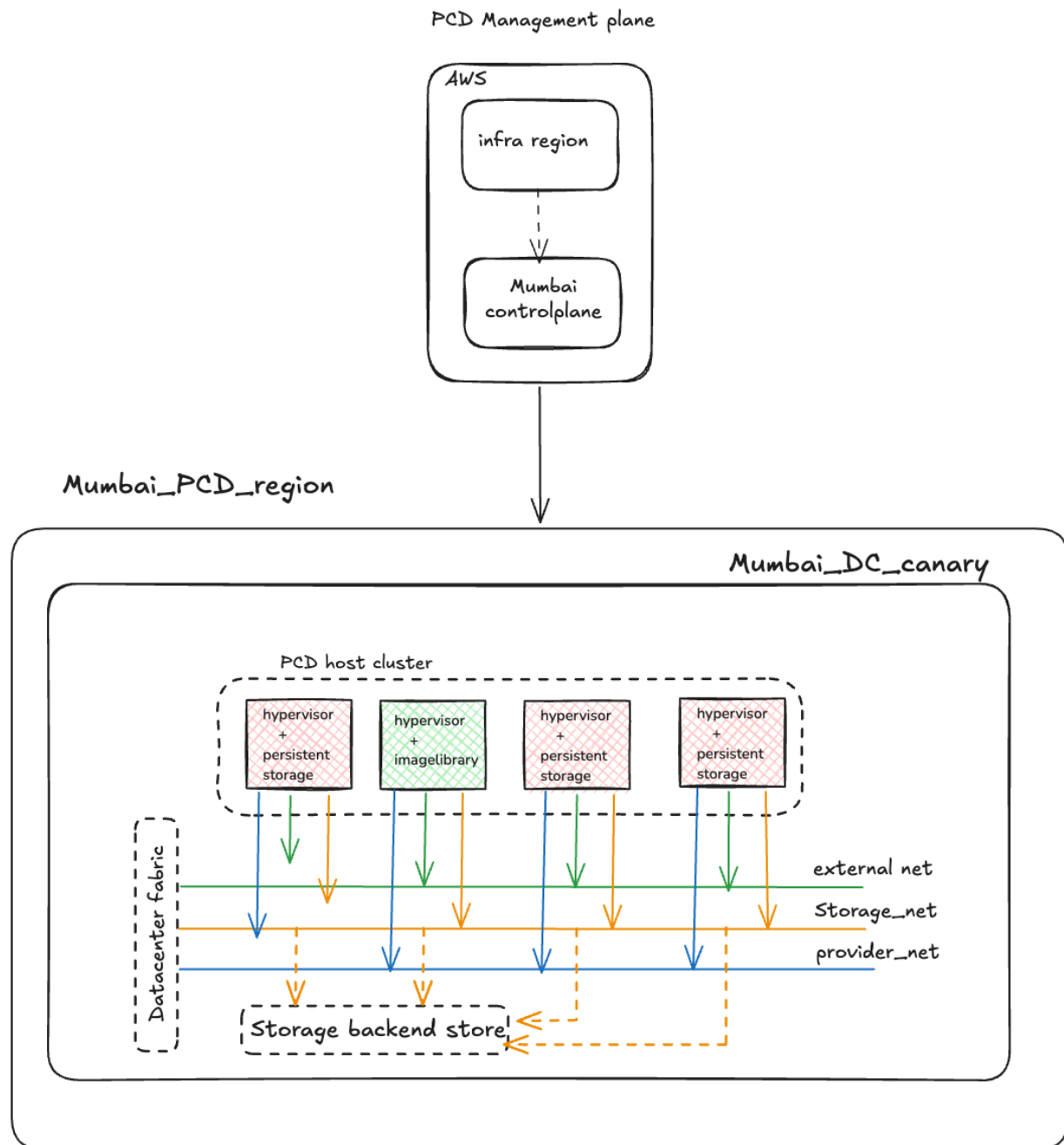
For enrolling bare metal nodes into the PCD DU regions at a large scale, a separate MaaS server would be deployed in the environment

Requirements:

Configuration	Details
Compute resource	• Virtual or baremetal host installed with u22.04 or ubuntu 24.04LTS • Minimum 4 Core 16GB RAM and 130GB disk
Network	• Single or multiple interface to access: • External internet • IPMI network of the baremetal hosts • A dedicated VLAN or native VLAN setting for PXE network access

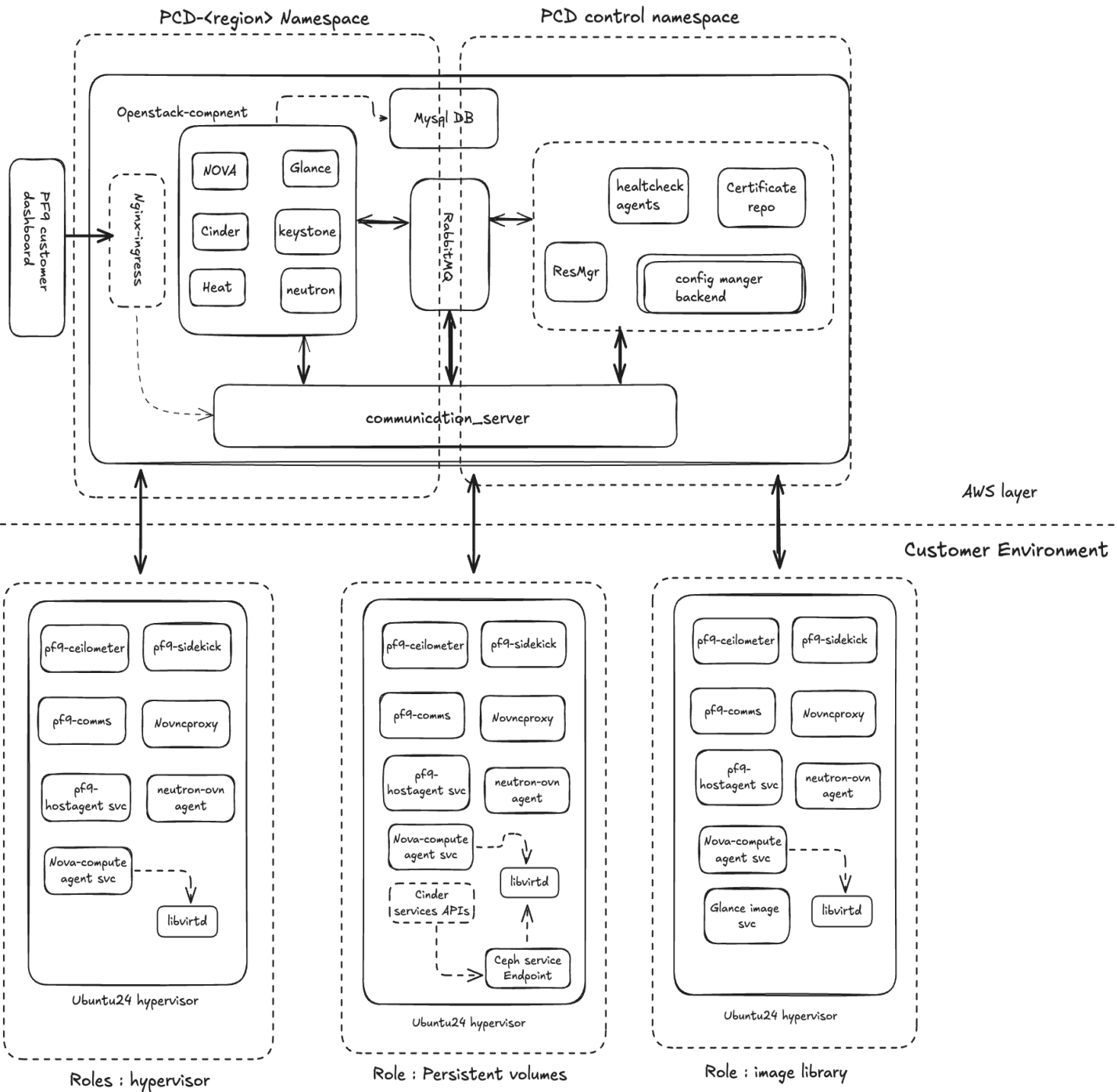
High-Level Solution Design (HLD)

Conceptual view of PCD deployment



Low-Level Solution Design (LLD)

Detailed logical Diagram of PCD deployment



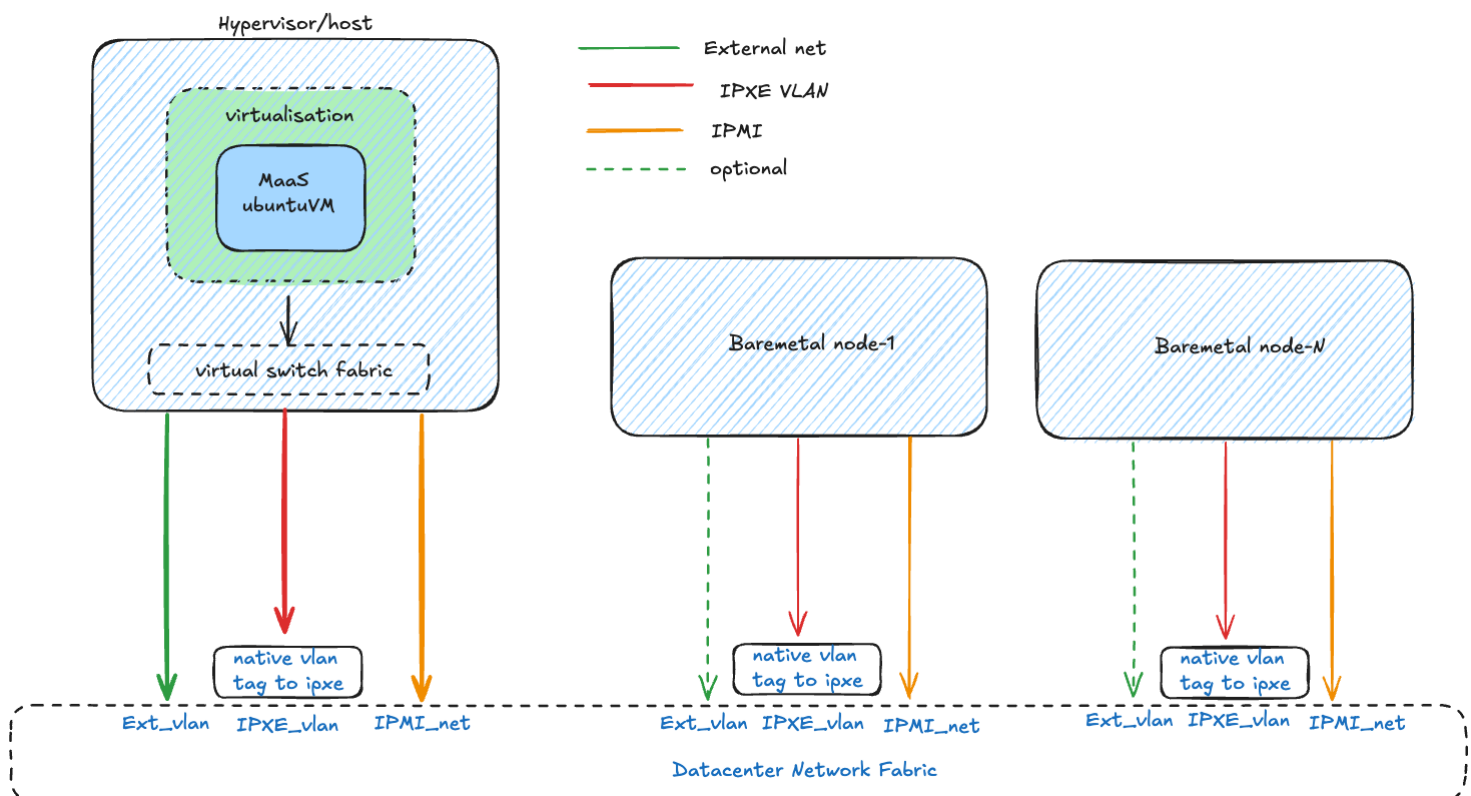
Integration Architecture

Setup MaaS service in the environment to configure baremetal hosts

For a large-scale re-imaging of baremetal nodes a MaaS server would be deployed which can either be a **VM residing in VMware or a baremetal host** depending on the requirements.

Pre-requisite

For successful deployment of MaaS and onboarding of host, the MaaS server needs the access to specific networks:



- **External network:** MaaS will use this network to connect with maas.io for syncing image repo which will be used for deploying the hosts.
- **iPXE network:** This should be an isolated network to which MaaS and the baremetal hosts to be re-imaged will be connected over. The Baremetal host NIC associated with this

network should have PXE boot service enabled. **MaaS will use this network to provide dhcp IP to the PXE hosts and images to deploy and configure the host.**

- **IPMI Network:** MaaS will have access to the IPMI network through which it would trigger the power up and power down operation needed during the deployment.

Install and configure

- Deploy a **ubuntu 24.04LTS** VM host with the pre-requisite configuration in place.
- Run the `maas_configure.sh` script which will setup the host with **maas 3.6.2** release version
- Post install, login to the UI to disable the network discovery.
- Configure the domain suffix.
- Run the image sync operation to ensure the latest ubuntu 24 LTS image is ready.
- Post installation access the web UI of maas and test for a successful commissioning operation of one of the target host to ensure the system is ready and functional

Setup and re-image hosts

Manual method:

- From the add hardware tab > select Add machine and fill in the:
 - Machine profile name
 - Domain
 - Architecture
 - Kernel
 - Power type
- Select the cloud-init section and copy the cloud-init details that will set up the hosts with the required ssh logins, network configuration with the needed IPs in place.
- Trigger the commissioning.
- Once the commissioning completes successfully, run the Deploy action to install the OS.

Automation: (for re-imaging and onboarding baremetal at scale)

- Login to the maas server over ssh.
- Download the latest pcd-maas scripts.
- Run through the configuration steps as provided in the repo readme
- Execute the script with the prepared list of host details in the CSV file

Onboarding of hosts and cluster creation

- Once the host is reachable over the network, follow the Add new hosts workflow to download the binaries and prepare the hosts.
- From the Portal UI create a cluster resource and enabled the VMHA DRR services for the cluster

Add Cluster



Create a new cluster to organize and manage hosts for workload deployment.

Cluster Configuration

Name *

? Hint

Name *

CPU Mode

? Hint

Default

Default



VM High Availability ?

This feature requires at least 2 hosts that are authorized as hypervisors.



Dynamic Resource Rebalancing ?

Dynamically rebalance workloads across hosts in the cluster.



GPU ?

Enable running GPU workloads on the cluster.

NOTE:

- VMHA and DRR are responsible for providing service equivalent to the *vmotion* and *DRS* functions in VMware.
- This can be enabled or disabled per host cluster

Configure PCD blueprint section to match the environment specification

Setup the cluster name DNS and the overlay segmentation options:

Cluster Blueprint

Cluster Blueprint allows you to design and configure your cluster to meet your intended use cases.

Name: rspc-local

1 Network Configuration

Cluster Network Parameters

⚠ Some configurations may not be changed while a hypervisor role is assigned.

DNS Domain Name * 🔍 Hint

sa-rspc.example.com

☒ Enable Distributed Virtual Routing (DVR)

☒ Enable Virtual Networking

Segmentation Technology *

IP Underlay for GENEVE Overlay

Geneve Tunnel ID Range * 🔍 Hint

5000-65000

NOTE:

- The DNS domain Name will be used for setting up FQDN for the VMs (hostnames)
- The Domain name set in the blueprint still does not guarantee that it will be used during VM FQDN creation. It will work only when coupled with a DNS service like designate.
- For external DNS/IPAM integration, the FQDN names from external service can still take precedence over this value.

NOTE: Once a resource is applied and consumed the fields in the Blueprint UI will become immutable (greyed out)

Setup host network profile for PCD cluster and VM overlay

Configure/create network profile with necessary labels and PCD traffic mapped to the required interface.

Host Configurations

[Learn more](#)

Hosts in a cluster are automatically configured for you based on a Host Configuration.

List each network interface on your hosts, and assign it system traffic if any, and/or a physical network label. You can later create Physical Networks using physical network labels.

Name this configuration:

[Delete Configuration](#)

Network Interface	Physical Network Label	Management	VM Console	Image Library I/O	Virtual Network Tunnels	Host Liveness Checks
 ens3	physnet1	☒	☒	☒	☒	☒
 ens4	physnetens4	☐	☐	☐	☐	☐

[+ Add Network Interface](#)

☐ Enable GPU Passthrough

Name this configuration:

[Delete Configuration](#)

Network Interface	Physical Network Label	Management	VM Console	Image Library I/O	Virtual Network Tunnels	Host Liveness Checks
 ens3	physnet2	☒	☒	☒	☒	☒

[+ Add Network Interface](#)

☐ Enable GPU Passthrough

[+ Add Host Configuration](#)

NOTE:

- The interface defined in this within the network profile will be used by PCD for communication (service and VM connections alike).
- For the interface to be considered and mapped within PCD, it would either need to have a label set or it should have one of the PCD traffic mapped on to it.
- Interfaces not part of this configuration section, can continue to exist and be used for other communications by the hosts while being part of the PCD cluster.

Block storage driver integration for cinder services

Setup the storage driver configuration in the blue print:

2 Persistent Storage Connectivity

Register Volume Types to enable the use of Persistent Storage Volumes. Supported infrastructure integrations include:



Storage Volume Types

[Learn more](#)

Volume Type

Volume Backend Configurations

– nfs_cinder

nfs_cinder

Driver: NFS

+

Add Configuration

– nfs_glance

nfs_glance

Driver: NFS

+

Add Configuration

[+ Add Volume Type](#)

NOTE:

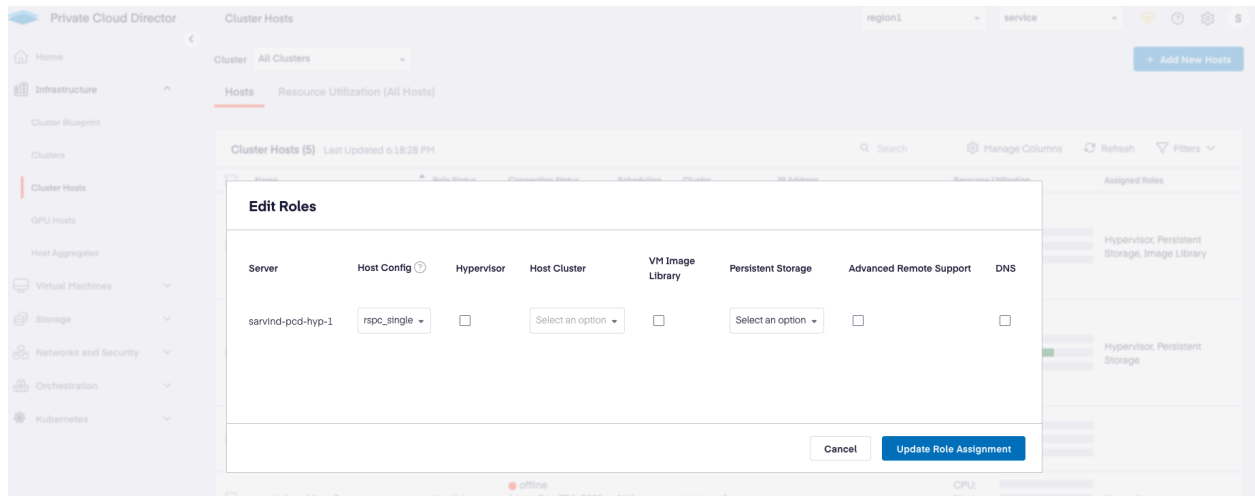
- Multiple hosts in a cluster can have the storage role/profile applied.
- Setting up multiple backend configurations for the same volume type will set up cinder to load balance and use all the defined backend within that volume type.
- Any changes in the backend configuration will get automatically pushed to the storage hosts in the cluster without any service restarts etc.

Network IPAM and DNS service integration

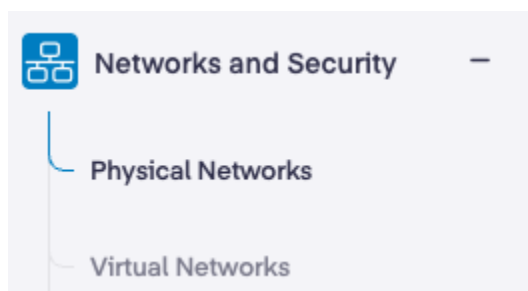
- There are no additional configurations to be defined within the PCD framework/blueprints.
- There are some adjustments that need to be set in the provider network configuration to ensure the external DNS service works consistently.

Onboarding the hosts with required roles into the pcd cluster

- As an administrator user, login into the portal and review the hosts under cluster hosts which are in the unauthorized state
- In conjunction with the current working and access to storage for the target nodes, select the host and use the edit role set the required role for the hosts within the specific cluster
- Every host with a role assigned will be part of a host cluster. Host cluster is constructed using the openstack availability zone and host aggregate methods.

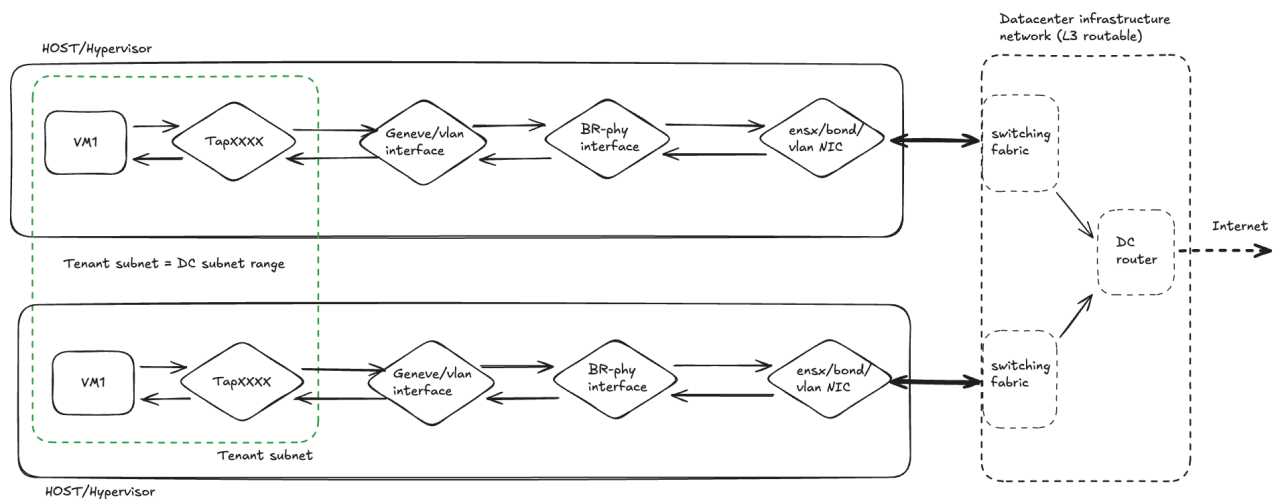


Create provider and overlay network configuration for the VM communication

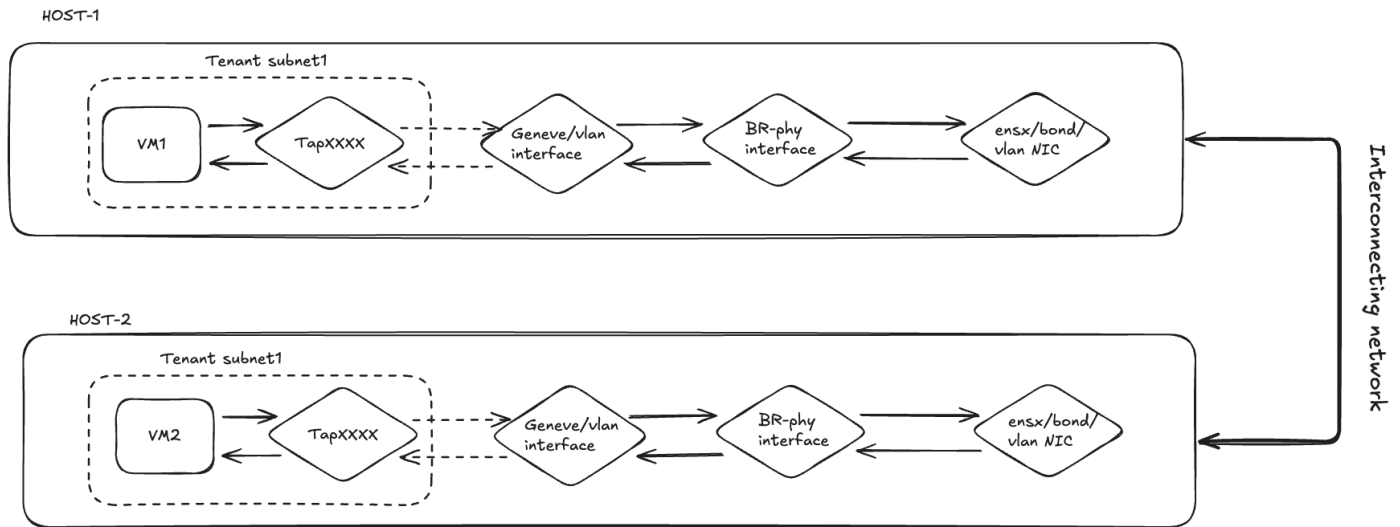


Depending on the use case configure/create a physical/provider network or a virtual network

Provider network general architecture view:



Virtual network architecture overview:



NOTE: It is recommended to have the port security enabled for the neutron ports. Hence application specific requirements need to be scoped during testing.

Setup Access integration of PCD login with AD IDP

- Ensure the EntraID application is created with basic SAML mapping and it has the respective group mapping created.
- In the SSO tab, setup the configuration and the attribute map xml file as needed
- Upon successful sync review the SAML group section and add/modify the respective group matching the actual AD group.

The screenshot shows the **SSO SAML Groups** configuration page. The **Enable SSO** toggle is turned on. Below the toggle, there is a description: "Enterprise Single Sign On supports SAML 2.0 identity integration for seamless access to your Private Cloud Director instance."

The configuration fields include:

- SSO Provider ***: A dropdown menu with the text "Select an option".
- Entity Endpoint for your SSO Provider**: A section containing:
 - Entity ID ***: A text input field with the placeholder "Entity ID *".
 - SAML Metadata URL ***: A text input field with the placeholder "SAML Metadata URL *".
- SSO Provider Attribute MAP in XML**: A section containing:
 - Default Attribute Map**: A table with one row and one column, containing the value "1".

Backup integration strategy

Solution implemented: Commvault

General procedure:

- Setup up and install commServe metadata server in the environment.
- Deploy media agent/access nodes which will be per PCD region (Single VM in this case)
- Add platform9 hypervisors as Openstack entities.
- Continue the process to discover the VMs and add the required VMs for backup operations.
- Trigger a restore action to validate the configuration.

Migration strategies for VM workload

Pre-requisite

- Ensure a provider network is available in the PCD which can be used for the vjailbreak tool to connect to Vcenter and perform the migrations of the VMs:
- Continue with the installation of Vjailbreak tool:
https://platform9.github.io/vjailbreak/introduction/getting_started/
- Fetch/compile the list of VMs targeted to be migrated to PCD.